



IIT MADRAS SOFTWARE ENGINEERING PROJECT



SMART BATCH PLATFORM

AI-DRIVEN OPERATIONS FOR LOCAL EDUCATION

TEAM NAME: HYPERLOOP CODERS

TEAM MEMBERS:

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|----------------------------|------|---|
| 1) SAMARTH S. | ---- | <i>Product Manager & Frontend Developer</i> |
| 2) DEEBHIKA KUMARAN | ---- | <i>Backend Developer & AI Implementation</i> |
| 3) VEDANT GOYAL | ---- | <i>Code Reviewer & Database Administrator</i> |
| 4) SANTHANU | ---- | <i>Code Reviewer & AI Integration work</i> |
| 5) SAMEER THAKUR | ---- | <i>Scrum Master & Code Tester</i> |

TEAM CODE: MAY2026-TEAM-025

START
PRESENTATION

WHO USES SMARTBATCH ?

One platform. Different needs. One connected ecosystem.

PRIMARY USERS

ADMIN & TUTORS (Centre Owners)

Profile:

- Run the tuition centre
- Manage daily operations
- Handle students & parents

Main Tasks:

- Fee & payment tracking
- Attendance
- Notes & homework
- Tests & announcements

Need:

- Less Manual work
- More focus on teaching

SECONDARY USERS

PARENTS & STUDENTS

Parents:

- Track child's attendance
- Check fee status
- View test performance
- Receive important alerts

Students:

- View notes and Resources
- Check timetable and tests
- Submit Homework
- Get help through AI chatbot

Goal:

- Easy access to everything
- less dependence on whatsapp

TERTIARY USERS

SUPPORT USERS

External Accountant:

- Access fee records
- View financial reports
- Check payment history

Technical team:

- Monitor system issues
- Handle bugs & errors
- Maintain platform reliability

Purpose:

- Support smooth operations
- Keep the system running

PROBLEM STATEMENT

01

ADMINS & TUTORS:

"Too much time goes into managing."

- Searching paper registers for fee dues.
- Remembering custom payment cycles.
- Manually calculating monthly earnings
- Repeated calls for holidays & updates

02

ADMINS & TUTORS:

"Teaching gets interrupted."

- Parents call to check if their child arrived
- Attendance has to be handled manually
- Wrong batch timings disturb classroom seating
- Constant communication takes time away from teaching

03

PARENTS:

"I just want to know my child is safe & on track."

- No easy way to confirm arrival
- Fee reminders can be confusing
- Important holiday updates get buried in WhatsApp groups
- Need to repeatedly contact the tutor

04

STUDENTS:

"Missing one class shouldn't mean chasing someone for notes."

- Depend on classmates for handwritten notes
- Notes are shared through distracting messaging apps
- Wrong batch timings disrupt their routine
- Holiday information can easily be missed

THE COMMON PROBLEM:

"Everything is scattered"

- Paper registers + WhatsApp + phone calls + manual calculations
- There is no single place to reliably manage:
Fees • Attendance • Schedules • Resources • Communication

05

OUR TECHNOLOGY STACK



FRONTEND

REACT 19 + VITE

- React Router 7 for navigation
- Tailwind CSS + shadcn/ui for UI
- Recharts for dashboards & analytics

ROLE:

Building the role-based interfaces for Tutors, Students & Parents.



BACKEND

FLASK + POSTGRESQL

- SQLAlchemy + Alembic
- JWT + Bcrypt authentication
- Supabase Storage for files
- APScheduler for background jobs
- Razorpay for payments

ROLE:

Handling APIs, user management, data, payments and core application logic.



AI / ML

DEEPPFACE + OPENCV + TENSORFLOW

- Facial recognition for attendance
- LiteLLM for AI-based test grading
- FastAPI AI microservice
- Pinecone + Gemini Embeddings RAG
- AI study assistant/chatbot

ROLE:

Powering attendance, grading & student assistance..

SMARTBATCH — KEY HIGHLIGHTS

CORE OPERATIONS

Separate dashboards and permissions for Admin, Teacher, Student & Parent.

Manage classes, subjects and assignments, with file-based homework submission.

Track fee dues and recurring payments with Razorpay integration and payment verification.

Send important updates and automated alerts for homework, marks, fees and payments.

Track marks, attendance and academic performance through visual dashboards.

SMART FEATURES

Upload a class photo and let facial recognition identify enrolled students and assist with attendance.

Students submit answer sheets → AI grades them against the answer key, including per-question scoring.

Students can ask questions about their course material through a RAG-powered AI assistant.

Teachers upload PDFs, notes and learning material; students access them directly from their dashboard.

Background scheduling and notifications reduce repetitive manual work for tutors and admins.



SmartBatch brings attendance, fees, learning, communication and AI-powered tools together in one simple platform for tuition centres.

THANK YOU FOR YOUR TIME

QUESTIONS?

